

Energy Transition: Opportunities and Challenges to the Pipeline Industry

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Abstract

The energy transition is gaining momentum due to the global consensus on addressing climate change. Policies and commitments to reduce greenhouse gas emissions have transformed the business landscape, with companies, including pipeline companies, needing to align their strategies with climate goals. Decarbonization policies, such as carbon markets and taxation systems, and increased scrutiny on compliance are driving change. The pipeline industry faces significant changes as the energy transition progresses. While oil and gas transportation will decrease, pipelines will be needed for low-carbon fuels and CO₂. This chapter assesses the opportunities and challenges for the pipeline industry related to the energy transition. The chapter shows the need for adaptation to the new business environment, which requires integrating new energy sources, developing hybrid grids, and engaging stakeholders. Challenges include establishing technical standards and certification systems for different energy

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sources and investing in technology for safety. Technological innovations are crucial for hydrogen pipeline safety and corrosion-resistant materials for CO₂ pipelines. Overall, the pipeline industry must adapt strategies, processes, and technologies to support the energy transition and ensure a smooth and efficient transition to low-carbon energy sources.

Keywords

Pipeline industry · Energy transition · Low-carbon fuels · CO₂ · Hydrogen · Hybrid grids · Technical standards · Certification systems · Technological innovations · Safety

Introduction

Energy transition is gaining momentum with the development of a growing political consensus regarding the challenges and implications of the climate change. An increasing number of governments and companies have committed themselves to energy transition, aiming to achieve a net zero emission targets until 2050 or 2060. Energy transition will certainly affect the way energy is produced and transported. Therefore, the pipeline industry will be directly affected by the changes in the energy sector policies and economics.

As energy transition progresses, the way energy is produced and transported will change radically. The decarbonization of the world economy requires the oil and gas production to be progressively replaced by low-carbon sources of energy. One important trend for decarbonization of the economy is the electrification of the economies, using renewable electricity sources. However, today only about 20% of the final energy consumption is covered by electricity. Therefore, it is unlikely electrification can replace all the energy demands covered today by oil and gas. Other liquid and gaseous low-carbon energy sources will be produced and transported by pipelines, which means that there is a future for pipelines in the transitioned energy systems.

Better understanding the implications of energy transition for pipeline industry, and the ways in which this industry should adapt to the challenges of decarbonization, is crucial to achieve net-zero carbon objectives. In particular, it is important to understand the role of pipelines in the energy transition. This chapter is dedicated to analyzing this question.

We can point out two initial answers to be further analyzed in the chapter. First, the pipeline industry will need to adapt their technologies and business models to the new business environment that energy transition policies and strategies create. Second, the ESG practices require that pipeline companies play their part in decarbonization. Therefore, pipeline companies must build and operate their assets to make them an enabler of the energy transition.

This chapter analyzes the opportunities and challenges for the pipeline industry related to the energy transition. The chapter is organized in three sections. First, the

perspectives for the energy transition are analyzed, highlighting the main drivers and the different scenarios for the energy transition. The second section is dedicated to the analysis of the opportunities for the pipeline industry related to the new energy sources. The third section points out the main challenges for the pipeline industry to become an enabler of the energy transition. Finally, the paper draws some conclusions regarding the importance of engaging the stakeholders of the pipeline industries on dealing with the challenges related to the energy transition.

Perspectives for Energy Transition

Before analyzing the role of pipeline companies in energy transition, this section will examine why energy transition policies have gained momentum and what are the perspectives for the energy transition in the next three decades. This section shows that the energy sector will change substantially in the next decades and pipeline companies have no other option than integrate the energy transition in their business strategies.

Why Now?

This energy transition can be defined as the structural changes in the world energy sector toward the low-carbon intensity sources of energy. The idea of an ongoing energy transition is not new. Since the oil shock in the 1973, energy policy has been oriented to reduce the world dependence on oil and gas, particularly from oil. Nevertheless, after the Paris Agreement, a new political consensus has developed regarding the importance of climate change and the role of fossil fuels emissions in the global warming. Climate scientists have managed to convince stakeholders that energy transition can limit global warming to 1.5 °C by 2050. As a result, a political consensus has emerged toward the idea that a rapid change in current world energy matrix must be a target for all stakeholders, in order to mitigate the climate changes and its consequences for the world economy.

This new political consensus has changed the purpose of political parties, government policies, the international cooperation, and companies strategic planning. Today it is almost impossible to not see climate engagement and energy transition as an important part of the stakeholder's agenda, including in the pipeline industry.

An important landmark of the new moment for the energy transition is the Green New Deal agreement in the European Union after Covid-19 pandemic in 2019. The EU decided to accelerate the energy transition, not only to align its energy policy with the climate ambitions, but also as a strategy to promote the economic recovery and development. An important dimension of the Green New Deal was the commitment of the EU with a Net Zero Emissions target by 2050. In order to reach this target, the European Commission intends to allocate 825 billion of euros for economic recovery by investing in wind, solar, green hydrogen, and energy efficiency.

Other important economies have followed the EU and committed themselves to the low-carbon trajectory. The United States not only have announced a target of net zero emission in 2050, but also have approved an Inflation Reduction Act which includes a US\$370 billion environmental package to combat global warming and encourage the adoption of renewable energies. China also approved a target of net zero emissions by 2060 and has made the renewable energy sources a priority for the energy investment in the next decades.

According to the OECD (2021), 52 countries representing around 70% of global carbon emissions have committed to net zero trajectories. Of these 52 countries, 17 countries have net zero emissions targets adopted in law. As government translates net zero target into laws, large multinational companies have no other option than commit themselves to a net zero emission targets. As a result, all major international oil companies (IOCs) have announced their commitment to the energy transition and net zero emissions trajectory.

The new political consensus described above is progressively being translated into decarbonization policies. One of the most important trend is the diffusion of carbon markets and taxation around the world. According to the World Bank (2022), 68 countries have implemented carbon markets or a carbon tax system and 68% of global emissions are already covered by market or carbon tax systems. Not only new countries are adopting this type of policy, but also, we see a progressive increase in emission restrictions, growing from scope 1 to 3 emissions.

Another important trend is an increasing scrutiny by government and other stakeholders regarding the companies' compliance with voluntary targets for business decarbonization. Therefore, it is fair to say that the business environment for the energy companies has changed radically in the last five years. Currently, energy companies must respond to stakeholders' pressure to align their strategy to the climate ambitions. In addition, the introduction of new regulations and standards is placing the carbon intensity at the forefront of business performance. Therefore, all companies, including from the pipeline industry, will need to measure, certificate, and mitigate their carbon emissions.

Scenarios for Net Zero Emissions

Although there is a growing political consensus regarding the net zero emissions targets, still there is a high level of uncertainty regarding how energy transition will take place. Several international institutions and large energy companies have published studies about energy transition scenarios (IRENA 2021). These scenarios are views on possible transition trajectories. As we should expect, these views vary significantly as many economic, technological, political, and social issues could affect the speed and the direction of energy transition.

IRENA (2021) has compared different energy transition scenarios from different institutions such as IEA, IPCC, Greenpeace, and energy companies. Energy transition scenarios vary in terms of the level of future energy intensity of the economy; the future level of participation of renewable energy sources; and the future level of

electrification of end uses. For example, the views on the share of renewables in 2050 range from 40% to 90%.

The level of electrification of the world economy by 2050 is very important for the pipeline industry. The higher the electrification rate, the lower the need for pipelines to transport liquid or gaseous sources of energy. IRENA (2021) have compared the share of the electricity in the energy supply in different energy transition scenario, showing a variation between 35% and 60%. As expected, the higher the share of renewable electricity in the final energy supply the lower the emissions. Nevertheless, it is clear that even in the most optimistic scenario for the rate of electrification, other renewable liquid and gaseous sources of energy will be necessary to be produced and transported by pipelines (Fig. 1).

The most notorious scenario for the net zero emissions in 2050 is the IEA scenario. This scenario presents a roadmap for the policies, technologies, and behavioral changes needed to achieve net zero emissions in just three decades. More than a forecast, the objective of the IEA Net Zero Emissions (NZE) scenario is to show the challenges, opportunities, and the necessary changes in the energy landscape in order to reach the energy transition (Fig. 2).

In the NZE Scenario, the share of coal, oil, and natural gas in the world energy matrix would reduce from the current 85% to about 20%. The fossil fuels would be replaced by renewable energy, especially solar and wind power, which would become the leading sources of electricity globally before 2030 and together will provide nearly 70% of global generation by 2050.

Fossil fuels remaining in 2050 are used in the petrochemical industry to produce non-energy goods, and will be associated to carbon capture, utilization, and storage

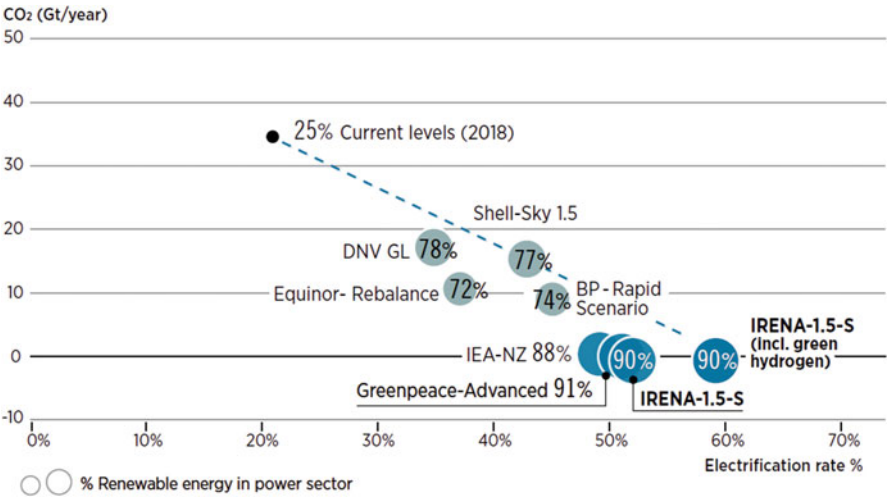


Fig. 1 CO₂ emissions versus electrification rates in various energy scenarios. (Source: IRENA (2021))

(CCUS). Coal demand declines by 90% in 2050, whereas oil and gas demands decrease by 75% and 55%, respectively.

A significant share of the expansion in renewables electricity capacity will be dedicated to produce hydrogen, which will be used directly or will be converted into low-carbon synthetic fuels. In the NZE Scenario, global hydrogen demand would increase from around 90 million tons (Mt) in 2020 to 530 Mt in 2050. Of this total, half would be used in heavy industry (mainly steel and chemicals production) and in the transport sector; and 30% would be converted into other hydrogen-based fuels, mainly ammonia for shipping and electricity generation, synthetic kerosene for aviation and synthetic methane blended into gas networks; and 17% is used in gas-fired power plants to balance increasing electricity generation from solar PV and wind and to provide seasonal storage. The NZE Scenario expects that hydrogen-based fuels account for 13% of global final energy demand in 2050 (Fig. 3).

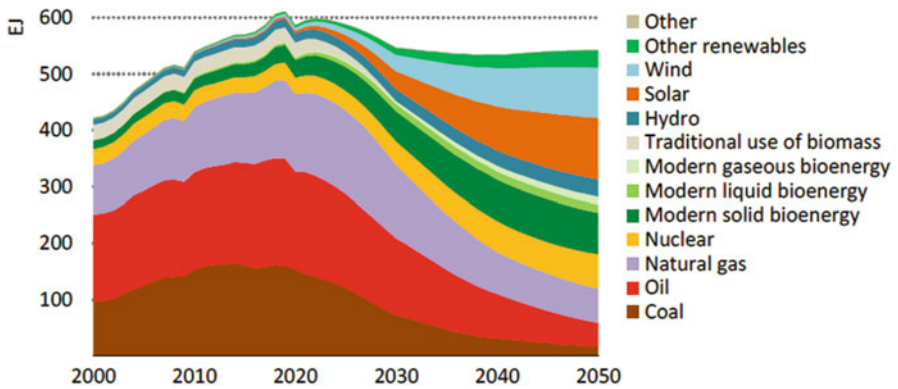
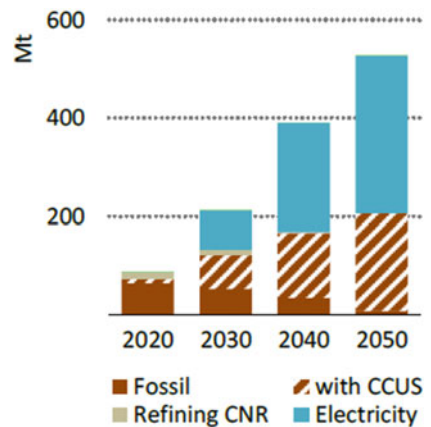


Fig. 2 The NZE Scenario from EIA. (Source: IEA (2021))

Fig. 3 Estimated evolution of hydrogen production in NZE Scenario. (Source: IEA (2021))



The use of CCUS (carbon capture, utilization, and storage) will be crucial to reach the NZE targets by 2050. CCUS can contribute to the transition to net zero in several ways. These include tackling emissions from existing energy assets, providing solutions in some of the sectors where emissions are hardest to reduce, such as cement, supporting the rapid scaling up of low-emissions hydrogen production, and enabling some CO₂ to be removed from the atmosphere.

According to the IEA's NZE Scenario, 625 Mt of CO₂ will be captured and stored from biofuel production. In addition, around half of the remaining fossil fuel use in 2050 will be in plants equipped with CCUS, allowing 3.5 Gt CO₂ emissions to be captured from fossil fuels in 2050. Around 925 bcm of natural gas will be converted to hydrogen with CCUS. In addition, around 470 Mtce of coal and 225 bcm of natural gas are used with CCUS in the electricity and industrial sectors, mainly to extend the operations of young facilities and reduce stranded assets.

Opportunities for the Pipeline Industry

The energy transition scenarios bring a clear message for the pipeline industry. The oil and gas production and transportation tend to reduce while new types of low-carbon fuels and CO₂ will be needed to be transported by pipelines. The stable technological and regulatory environment that pipelines industry has been operating in the last decades will be replaced by a business environment with fast changing demand, technologies, and regulations.

Although, energy transition could represent a challenge for part of the existing oil and gas pipeline infrastructure, pipelines will remain critically important for the energies of the future, such as biofuels, hydrogen, and CO₂ sequestration and storage. In addition, the existing pipelines will be used to transport fuels with different technical specifications. In this new business environment, the pipeline industry will need to manage a diversity of technological options, while ensuring energy security is maintained, considering the realities of both a methane and hydrogen-based economy.

This section points out the main opportunities for the pipeline industry related to the energy transition. Transformations in the pipeline industry are both related to the adaptation of current pipeline infrastructure to new gas specifications and to the development of new pipeline technologies and networks to new energy sources. Therefore, the pipeline industry tends to evolve to a hybrid energy system.

Decarbonization of the Natural Gas Industry

It is important to mention that the replacement of natural gas by other types of sustainable gaseous fuels tends to be a gradual process. One strategy for the natural gas industry decarbonization is to gradually blend sustainable fuels, such as biomethane, green hydrogen, and synthetic gas, in the existing natural gas networks. Another strategy is the development of new pipelines dedicated to new sources of

production could be a competitor for the hydrogen transportation by pipelines. However, the most efficient and competitive supply sources are not necessarily located close to the demand. A hydrogen pipeline network creates the opportunity for reducing the cost and make hydrogen more accessible to the end consumers.

As the demand for hydrogen increases, the development of dedicated hydrogen pipeline systems tends to become economically feasible. At this point, the role of pipelines companies will be to assess the technology readiness and to evaluate the feasibility of converting existing pipeline systems to accommodate 100% hydrogen.

CCUS

The development of pipelines dedicated to CO₂ transportation represents another important opportunity for the pipeline industry. CO₂ pipelines have been in service for over 30 years (Barrie et al. 2005; Furley and Carvalho 2007). Currently, there are approximately 8000 km of CO₂ pipelines globally (Lu et al. 2020). Most of the existing CO₂ pipelines are related to enhanced oil and gas recovery and are located in the USA. The CO₂ pipelines in the USA have been built in the twentieth century. More recently, several CO₂ pipelines have been built in the Brazilian offshore. The associated gas produced in the Brazilian Presalt area has a high level of CO₂ contamination. In order to produce natural gas from Presalt fields, Petrobras separates CO₂ from the natural gas in the platforms and transports CO₂ through pipelines to the injection wells.

Energy transition will require the development of a very large infrastructure of pipelines to allow CO₂ transportation from the places of sequestration to the places of re-use and/or storage. Figure 5 presents a schematic view of the CO₂ transport system via pipelines.

According to Imperial (2021), the CCS global portfolio of projects under development is growing rapidly. This portfolio increased from 75 Mtpa at the end of 2020 to 111 Mtpa in September 2021. In 2001, there were 28 projects in operation and under development globally including pipelines for CO₂ transportation. The DNV ETO and the IEA forecast CCUS demand in 2050 will be almost 100 times larger today. The IEA NZE scenario estimated that the annual investment in CO₂ pipelines and hydrogen enabling infrastructure could increase from USD 1 billion today to around USD 40 billion in 2030.

Figure 6 presents a schematic structure of CCUS technology. Pipelines will have a crucial role for the technical and economic viability of CCUS, given the high volumes of CO₂ needed to be captured and the increasing distances between the point of capture and storage. The first CCUS projects are being proposed at CCUS hubs, to increase the scale of CO₂ capture and to reduce the cost of its transportation (Global CCS Institute 2016). The most attractive CCUS hubs will be the ones where a significant amount of CCUS can be captured near the sites available for underground storage. Nevertheless, as the price of CO₂ credits increases and the CCUS technology evolves, the economics of CO₂ transportation in longer distances will increase, and so the demand for CO₂ pipelines.

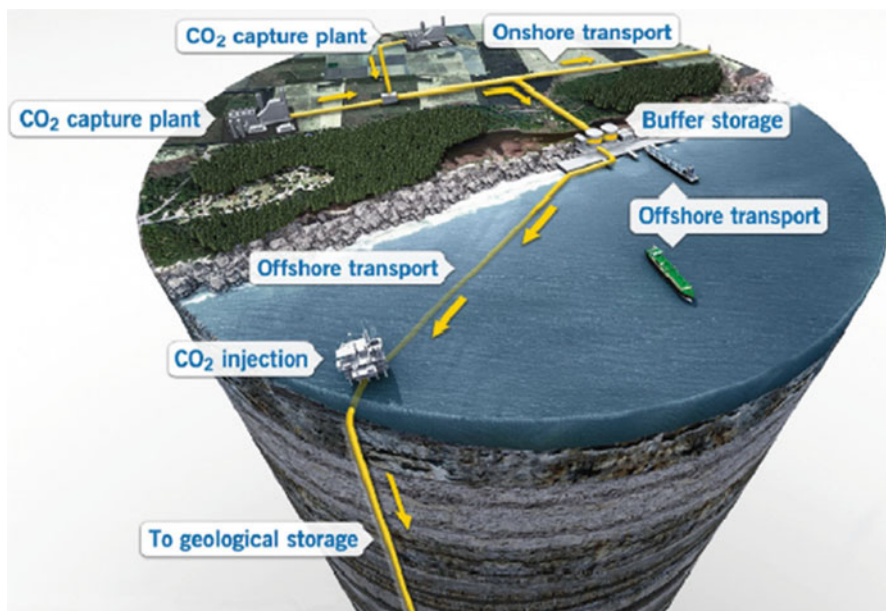


Fig. 5 CO₂ transport system. (Source: Global CCS Institute 2019)

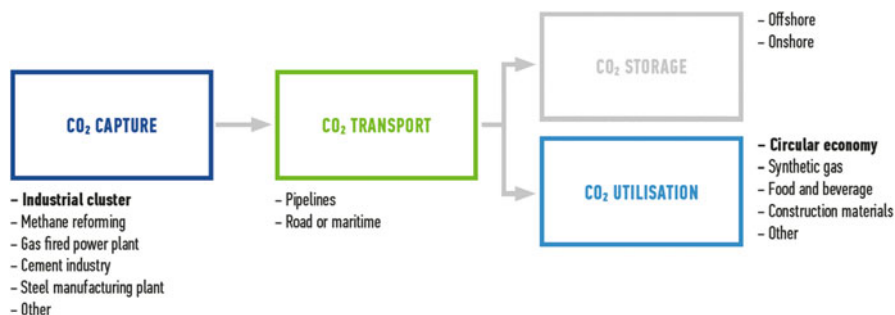


Fig. 6 CO₂ capture, transportation, and storage. (Source: ENTSG (2022))

It is important to mention that the technology of CO₂ separation from the air (direct air capture – DAC) can be an important competitor of CO₂ pipelines. This technology extracts CO₂ directly from the atmosphere, either by liquid or solid DAC technologies (IEA 2022). Once the CO₂ is separated, it can be permanently stored in deep geological formations or it can be used, for example, in food processing or combined with hydrogen to produce synthetic fuels. This technology has the advantage of being installed near the storage site, therefore avoiding the CO₂ transportation through long distances. If economically feasible, DAC technologies can be an important tool to allow carbon emissions compensation and even to achieve negative emissions.

Challenges for the Pipeline Industry

In order to take advantage of the opportunity described in session 1.3, pipeline companies will need to face the challenges related to these opportunities. One important challenge is to avoid stranded assets related to the market fragmentation, as hydrogen and biomethane usage develops. The pipeline sector needs to timely engage with the stakeholders involved with energy planning and regulation to avoid that energy transition results in energy networks disintegration. This engagement should aim to allow the establishment of a hybrid gas grid to maintain the role of current pipeline system in energy supply.

For an integrated and hybrid gas system to emerge, several obstacles will be needed to be tackled. Firstly, it will be crucial to create a credible system to identify and certify the different types of gases to be injected in the network. This system is needed to ensure that biomethane, hydrogen, and other renewable or low-carbon gases can be tracked from production and/or import to consumption. A proper system of certification will allow to sell separately the gas commodity, the CO₂ certificates, and the Guarantee of Origin certificate. This separation would allow the reduction of transaction costs and create a liquid market for new types of low-carbon gases.

Another important obstacle to be dealt with is the need for the new technical standards regarding the hydrogen-methane blends, as well as the revision of the end-use applications' technical standards. The process of standard revision is complex and lengthy, as all stakeholders should be involved. In the case of the EU this is even more complicated as the process needs to consider each member state. Therefore, it will be crucial to accelerate the discussions regarding technical standards creation and revision in order to meet the NZE targets and to avoid the pipeline disintegration and its consequences related to the rising of stranded assets.

Poudineh (2022) calls the attention to the importance of regulation to reduce the risks and, consequently, the cost of hydrogen networks, since currently this activity is mostly unregulated. Therefore, this author emphasizes the importance of developing an appropriate regulatory framework, similar to the current natural gas networks. An important dimension of the future hydrogen network's regulatory framework is the integration of existing commercially driven investments in pipelines into future regulated hydrogen network infrastructures.

ENTSOG (2022) recommends that a coordinated EU approach for managing the changes in gas compositions across Europe. It should set the basis for the revision of gas quality standards and for end-user appliances and CNG vehicles. Development of hydrogen readiness targets could be considered, including analysis of costs and benefits of such a transformation. Measuring and sharing of data of the gas composition in both the distribution and transportation networks will also be crucial.

Another important obstacle to be dealt with is the development of technologies needed to cope with new types of energy sources. The IEA has estimated that only about 50% of the technologies required to reach the NZE Scenario in 2050 are available in the market today. Therefore, the technological innovations represent an

important challenge for all segments of energy industry, and this is not different for the pipeline industry.

Technological Challenges Related to H2 Pipelines

To develop a hydrogen pipeline network, it will be necessary to tackle different new technical issues such as blending, conversion, separation, flow management, and so on. Therefore, pipeline companies will have to invest to develop new technologies, such as new pipeline material, smart metering, gas quality detection, certification and data sharing, and data provision by and between the pipeline companies and consumer appliances.

New business models will be needed to make possible for the pipelines companies to deal with the energies of the future. These new business models will involve changing regulations to allow existing gas pipeline companies to transport other sources of gaseous energy, as well as to supply new types of services enabling quality management, energy conversion, and interoperability of different gases (ENTSOG 2022). (See also Poudineh (2022) for a view on the role of energy networks in the transition of energy systems.)

One of the most important technological challenges for the reaching NZE targets for 2050 is the development of a cheap and safe way to transport hydrogen. Hydrogen is a gas much lighter than natural gas, with a high propensity of ignition. Therefore, hydrogen transportation by pipelines presents several safety and other operational challenges.

ENTSOG (2022) shows that it is possible to blend low concentrations of hydrogen in the gas network without significantly increasing risks associated with utilization of the gas blend in end-use devices, overall public safety, or the durability and integrity of the existing natural gas pipeline network. Nevertheless, the definition of the level of hydrogen in the blend would require an extensive study to determine the optimal hydrogen level. The introduction of hydrogen blends would certainly add costs to the pipeline system due to the necessary technical adaptations and the modifications of existing pipeline monitoring and maintenance practices.

GONDAL (2019) shows that each element of the natural gas infrastructure has a varying degree of acceptability to hydrogen concentration. Compressors are the determinant elements since they accept only up to 10% of hydrogen blending. The gas distribution network and storage elements tolerate up to 50% concentration of hydrogen. End-use appliances have a tolerant range of 20 to 50%. This study has also analyzed the effect of hydrogen blends on gas quality, showing that it is possible to introduce 2% hydrogen in the distribution network maintaining the gas specifications. Nevertheless, a 10% hydrogen blending would affect the calorific value of the supplied fuel requiring changes in the gas specifications.

Melaina et al. (2013) show that impact of hydrogen blends on industrial facilities must be addressed on a case-by-case basis. This study shows that hydrogen blending likely would begin at very low concentrations and then increase gradually over time, as required modifications for higher concentrations are addressed.

The impact of hydrogen blends to the gas losses is a very important technical issue. Since hydrogen is more mobile than methane, hydrogen losses can be a complex technical issue. Permeation rates for hydrogen are about four to five times faster than for methane in typical polymer pipes used in the gas distribution system. Leakage measurements distribution pipeline systems suggest that the volume leakage rate for hydrogen is about a factor of 3 higher than that for natural gas. Melaina et al. (2013) have shown that blending low concentrations of hydrogen in the natural gas networks does not increase significantly the safety risks. However, higher concentrations (more than 50% hydrogen) result in a significant increase in overall safety risks in the gas network. It is important to mention that this does not apply to new, dedicated hydrogen pipelines carrying pure hydrogen, which would be designed and managed differently according to the hydrogen safety specifications.

New metering technologies and digitalization will be crucial to allow the new hybrid pipeline systems. Hydrogen blends can influence the accuracy of existing gas meters. Therefore, the introduction of hydrogen blends would require meters recalibration. Precise and quick exchange of gas quality data will be essential for operating the gas systems, providing information to end-user appliances as well as for ensuring fair and transparent billing processes.

Regarding the dedicated hydrogen pipelines, one of the most important challenges is the safety risks associated with uncontrolled gas leaks. Witkowski et al. (2017) have shown that uncontrollable hydrogen release due to damage in the pipelines poses a potential hazard to humans and the surroundings. For a pipeline diameter of 250 mm, the zones with a fatal risk on humans extend from the location of the pipeline damage over approximately 120 m. The zones presenting a hazard to human health and life will depend, among others, on the hydrogen pressure and the size of the pipeline damage.

Yang et al. (2021) make a detailed review of the safety issues of hydrogen pipelines. This paper has analyzed 120 hydrogen safety incidents that occurred from 1999 to 2019 and produced statistical data on the locations, facilities, possible causes, and losses. This data has shown that failures of pipes/valves/filters in the hydrogen system were notably the most common cause of hydrogen-related incident. Therefore, the development of pure hydrogen pipeline will require an important improvement of the risk monitoring and prevention strategies and protocols.

Li et al. (2022) have made a detailed review of the research progress on hydrogen leak diffusion characteristics, leak spontaneous combustion mechanisms, and material hydrogen damage mechanisms. This analysis shows that there are still several open research questions regarding safety issues related to hydrogen leakage, such as: exploring the expansion and dilution law of hydrogen clouds after liquid hydrogen flooding; further studying the spontaneous combustion mechanism of leaked hydrogen and the interaction between mechanisms; and investigating the synergistic damage effect of hydrogen and other components on materials. This paper suggests that an important research effort will be necessary to create the adequate technological knowledge in utilization of pure hydrogen pipelines.

Fuel Cells and Hydrogen (2019) proposes an overview of the impacts of hydrogen in the gas distribution pipelines and appliances (Fig. 7).

	H ₂ -methane blending	Pure H ₂ networks
Distribution infrastructure	Blending of gaseous H ₂ into existing natural gas pipelines is possible up to a concentration of ~5-15%¹ – modifications to existing pipeline monitoring and maintenance practices necessary to ensure safety	Retrofitting or replacement of existing steel pipelines to noncorrosive and nonpermeable materials (e.g., polyethylene, fiber-reinforced polymer pipelines) and leakage control is required for the transportation of pure gaseous H ₂
Gas heating and cooking appliances	Utilization of H ₂ -methane blending in existing end user appliances is possible up to a concentration of 5-20%¹ , when calorific values are kept within tolerance bands; research even suggests 30% is possible , allowing for appropriate margin of safety	Conversion or replacement of end user appliances [gas boilers, hot water tanks, gas cookers] required

Fig. 7 Hydrogen pathways to decarbonization. (Source: Fuel Cells and Hydrogen (2019))

Technological Challenges Related to CO₂ Pipelines

There are important technological, operational, economic, and regulatory challenges for the transportation of CO₂ at the scale necessary for the NZE scenarios. The high cost of CO₂ transportation from the place of capture to the place of storage represents an important challenge to the diffusion of CCS. One important strategy for reducing the cost of CO₂ transportation is to explore economies of scale, by transporting CO₂ captured from different industries in CCS hubs. However, the level of CO₂ purity varies according to the source of CO₂, which creates technical challenges to large integrated CO₂ pipelines (Gao & Liu 2017; Onyebuchi et al. 2018).

Lu et al. (2020) have also reviewed the application of CO₂ pipelines considering pipeline design, process, risk and safety, standard and specification. This paper has shown that the specifications and standards for carbon dioxide pipelines are not yet developed and that the techno-economic framework and integrity management system are still immature. The paper's review of the literature has shown that impurities in CO₂ have a great impact on corrosion of CO₂ pipelines. In addition, this review pointed out that the level of knowledge concerning ductile fracture and fracture control methods for CO₂ pipeline is still insufficient.

In the field of CO₂ pipeline material research, the study of failure mechanisms and the development of corrosion-resistant and high-grade materials are also a major challenge. (See also Bui et al. (2018) for an extensive assessment of the challenges for the CCUS development, including the CO₂ pipeline transportation.) In addition, the risk assessment and safety control of CO₂ pipeline are still to be developed. Based on this analysis, Lu et al. (2020) point out the future research directions for CO₂ pipelines (Table 1).

Conclusions

The analysis of the energy transition scenarios gives a clear message for the pipeline industry: the oil and gas production and transportation tend to reduce overtime, while new types of low-carbon fuels and CO₂ will be needed to be transported by pipelines. It is also clear that the stable technological and regulatory environment

Table 1 Future research directions of CO₂ pipelines

Research area	Future directions
Pipeline design	Effect of impurities on phase equilibrium
	Techno-economic framework for different CO ₂ pipelines (Onyebuchi et al. 2018)
	Optimization of transport scheme, especially from the perspective of energy consumption and safety
Pipeline material	Ductile fracture index
	Corrosion mechanism of CO ₂ pipeline in different environments
	Development of high-strength, corrosion-resistant steel
Pipeline construction	New construction technology, such as trenchless installation method
	Submarine pipeline construction
Safety and maintenance	Risk assessment system and method for CO ₂ pipeline
	CO ₂ leakage and diffusion mechanism
	Detection, monitoring, and emergency repair technologies for CO ₂ pipeline
Management	Formulation of specifications and standards specifically for CO ₂ pipelines
	Application of intelligent pipeline in transport deployment and monitoring
	Application of digital twin technology in safely management and early warning of CO ₂ pipeline
	Establishment of CO ₂ pipeline integrity system

Source: Lu et al. (2020)

that pipelines industry has been operating in the last decades will be replaced by a new business environment with fast changing demand, technologies, and regulations.

Given the profound changes in the business environment in the next decades, the pipeline industry must adapt their business strategies, processes, and technologies to the energy transition.

This chapter has pointed several opportunities and challenges to the pipeline industry. The integration of the new energy sources to the existing pipeline networks represents an effective strategy to avoid energy networks disintegration and stranded assets. Therefore, the establishment of a hybrid gas grid would allow to maintain the role of current pipeline system in energy supply. Nevertheless, the pipeline sector needs to timely engage with the stakeholders involved with energy planning and regulation to develop the concept of hybrid energy grids.

One of the main challenges to accelerate the adaptation of existing pipelines to the energy transition business environment is the development of new technical standards and a system of certification for the different types of energy sources being transported in the hybrid pipeline systems. The blending of hydrogen will require the revision of the pipeline and the end-use applications' technical standards. The development of standards for the CO₂ transportation is also an important task to allow the development of this business segment. Finally, a credible system to identify and certify the different types of gases to be injected in the network is needed to ensure that biomethane, hydrogen, and other renewable or low-carbon gases can be tracked from production and/or import to consumption.

Finally, this chapter highlighted the role of technological innovations in overcoming various technical obstacles associated with the construction and safe operation of hydrogen and CO₂ pipelines. In the case of hydrogen, pipeline companies will have to invest in new technology to improve the safety levels. This chapter has shown that it is necessary to improve the knowledge base regarding hydrogen pipeline safety, as well as to invest in new monitoring and maintenance protocols and practices. In the case of CO₂, technological innovations are crucial to develop corrosion-resistant and high-grade materials. Similarly to hydrogen, the study of failure mechanisms is very important to improve the risk assessment and safety control of CO₂ pipelines.

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